

The Roots of Problems

W.R. Casper

Problem Solving Seminar

Department of Mathematics
California State University Fullerton

September 26, 2025

Roots of polynomials

A **root** of a polynomial $f(x)$ is a solution of the equation $f(x) = 0$.

Roots of polynomials

A **root** of a polynomial $f(x)$ is a solution of the equation $f(x) = 0$.

TYPES OF QUESTIONS:

Roots of polynomials

A **root** of a polynomial $f(x)$ is a solution of the equation $f(x) = 0$.

TYPES OF QUESTIONS:

- where are the roots? (interval, region of complex plane, ...)

Roots of polynomials

A **root** of a polynomial $f(x)$ is a solution of the equation $f(x) = 0$.

TYPES OF QUESTIONS:

- where are the roots? (interval, region of complex plane, ...)
- how many real roots? (total, or in a certain interval)

Roots of polynomials

A **root** of a polynomial $f(x)$ is a solution of the equation $f(x) = 0$.

TYPES OF QUESTIONS:

- where are the roots? (interval, region of complex plane, ...)
- how many real roots? (total, or in a certain interval)
- given coefficients, evaluate an expression involving the roots

Roots of polynomials

A **root** of a polynomial $f(x)$ is a solution of the equation $f(x) = 0$.

TYPES OF QUESTIONS:

- where are the roots? (interval, region of complex plane, ...)
- how many real roots? (total, or in a certain interval)
- given coefficients, evaluate an expression involving the roots
- given the roots, evaluate an expression involving the coefficients

Warm-up problem

Problem

Let r_1, r_2, r_3 , and r_4 be the roots of the polynomial

$$f(x) = 5x^4 - 7x^3 + x^2 - x + 9.$$

Calculate

$$r_1 + r_2 + r_3 + r_4$$

Warm-up problem

Problem

Let r_1, r_2, r_3 , and r_4 be the roots of the polynomial

$$f(x) = 5x^4 - 7x^3 + x^2 - x + 9.$$

Calculate

$$r_1 + r_2 + r_3 + r_4$$

Solution

Warm-up problem

Problem

Let r_1, r_2, r_3 , and r_4 be the roots of the polynomial

$$f(x) = 5x^4 - 7x^3 + x^2 - x + 9.$$

Calculate

$$r_1 + r_2 + r_3 + r_4$$

Solution

$$f(x) = 5(x - r_1)(x - r_2)(x - r_3)(x - r_4)$$

Warm-up problem

Problem

Let r_1, r_2, r_3 , and r_4 be the roots of the polynomial

$$f(x) = 5x^4 - 7x^3 + x^2 - x + 9.$$

Calculate

$$r_1 + r_2 + r_3 + r_4$$

Solution

$$f(x) = 5(x - r_1)(x - r_2)(x - r_3)(x - r_4)$$

$$\begin{aligned} f(x) &= 5x^4 - 5x^3(r_1 + r_2 + r_3 + r_4) \\ &\quad + 5x^2(r_1r_2 + r_1r_3 + r_2r_4 + r_2r_3 + r_2r_4 + r_3r_4) \\ &\quad - 5x(r_1r_2r_3 + r_1r_2r_4 + r_1r_3r_4 + r_2r_3r_4) + 5r_1r_2r_3r_4. \end{aligned}$$

Warm-up problem

Problem

Let r_1, r_2, r_3 , and r_4 be the roots of the polynomial

$$f(x) = 5x^4 - 7x^3 + x^2 - x + 9.$$

Calculate

$$r_1 + r_2 + r_3 + r_4$$

Solution

$$f(x) = 5(x - r_1)(x - r_2)(x - r_3)(x - r_4)$$

$$\begin{aligned} f(x) &= 5x^4 - 5x^3(r_1 + r_2 + r_3 + r_4) \\ &\quad + 5x^2(r_1r_2 + r_1r_3 + r_2r_4 + r_2r_3 + r_2r_4 + r_3r_4) \\ &\quad - 5x(r_1r_2r_3 + r_1r_2r_4 + r_1r_3r_4 + r_2r_3r_4) + 5r_1r_2r_3r_4. \end{aligned}$$

$$r_1 + r_2 + r_3 + r_4 = \frac{7}{5}.$$

Warm-up problem

Problem

Let r_1, r_2, r_3 , and r_4 be the roots of the polynomial

$$f(x) = 5x^4 - 7x^3 + x^2 - x + 9.$$

Calculate

$$r_1 r_2 r_3 r_4$$

Warm-up problem

Problem

Let r_1, r_2, r_3 , and r_4 be the roots of the polynomial

$$f(x) = 5x^4 - 7x^3 + x^2 - x + 9.$$

Calculate

$$r_1 r_2 r_3 r_4$$

Solution

Warm-up problem

Problem

Let r_1, r_2, r_3 , and r_4 be the roots of the polynomial

$$f(x) = 5x^4 - 7x^3 + x^2 - x + 9.$$

Calculate

$$r_1 r_2 r_3 r_4$$

Solution

$$f(x) = 5(x - r_1)(x - r_2)(x - r_3)(x - r_4)$$

Warm-up problem

Problem

Let r_1, r_2, r_3 , and r_4 be the roots of the polynomial

$$f(x) = 5x^4 - 7x^3 + x^2 - x + 9.$$

Calculate

$$r_1 r_2 r_3 r_4$$

Solution

$$f(x) = 5(x - r_1)(x - r_2)(x - r_3)(x - r_4)$$

$$\begin{aligned} f(x) &= 5x^4 - 5x^3(r_1 + r_2 + r_3 + r_4) \\ &\quad + 5x^2(r_1 r_2 + r_1 r_3 + r_2 r_4 + r_2 r_3 + r_2 r_4 + r_3 r_4) \\ &\quad - 5x(r_1 r_2 r_3 + r_1 r_2 r_4 + r_1 r_3 r_4 + r_2 r_3 r_4) + 5r_1 r_2 r_3 r_4. \end{aligned}$$

Warm-up problem

Problem

Let r_1, r_2, r_3 , and r_4 be the roots of the polynomial

$$f(x) = 5x^4 - 7x^3 + x^2 - x + 9.$$

Calculate

$$r_1 r_2 r_3 r_4$$

Solution

$$f(x) = 5(x - r_1)(x - r_2)(x - r_3)(x - r_4)$$

$$\begin{aligned} f(x) &= 5x^4 - 5x^3(r_1 + r_2 + r_3 + r_4) \\ &\quad + 5x^2(r_1 r_2 + r_1 r_3 + r_2 r_4 + r_2 r_3 + r_2 r_4 + r_3 r_4) \\ &\quad - 5x(r_1 r_2 r_3 + r_1 r_2 r_4 + r_1 r_3 r_4 + r_2 r_3 r_4) + 5r_1 r_2 r_3 r_4. \end{aligned}$$

$$\text{Therefore } r_1 r_2 r_3 r_4 = \frac{9}{5}.$$

Warm-up problem

Problem

Let r_1, r_2, r_3 , and r_4 be the roots of the polynomial

$$f(x) = 5x^4 - 7x^3 + x^2 - x + 9.$$

Calculate

$$\frac{1}{r_1 + 2} + \frac{1}{r_2 + 2} + \frac{1}{r_3 + 2} + \frac{1}{r_4 + 2}$$

Warm-up problem

Solution

Warm-up problem

Solution

$$f(x) = 5(x - r_1)(x - r_2)(x - r_3)(x - r_4)$$

Warm-up problem

Solution

$$f(x) = 5(x - r_1)(x - r_2)(x - r_3)(x - r_4)$$

$$f(-2) = 5(-2 - r_1)(-2 - r_2)(-2 - r_3)(-2 - r_4)$$

Warm-up problem

Solution

$$f(x) = 5(x - r_1)(x - r_2)(x - r_3)(x - r_4)$$

$$f(-2) = 5(-2 - r_1)(-2 - r_2)(-2 - r_3)(-2 - r_4)$$

$$\frac{5}{f(-2)} = \frac{1}{r_1 + 2} \frac{1}{r_2 + 2} \frac{1}{r_3 + 2} \frac{1}{r_4 + 2} = \frac{5}{151}$$

Problem

Show that the polynomial

$$f(x) = x^3 + 4x + 1$$

has at least one real root.

Warm-up problem

Problem

Show that the polynomial

$$f(x) = x^3 + 4x + 1$$

has at least one real root.

Solution

$f(1) = 6$ and $f(-1) = -4$, so the **Intermediate Value Theorem** says that there is a value x between 6 and -1 with $f(x) = 0$.

Warm-up problem

Problem

Show that the polynomial

$$f(x) = x^3 + 4x + 1$$

has at exactly one real root.

Warm-up problem

Problem

Show that the polynomial

$$f(x) = x^3 + 4x + 1$$

has at exactly one real root.

Solution

Note that $f'(x) = 3x^2 + 4 > 0$.

Warm-up problem

Problem

Show that the polynomial

$$f(x) = x^3 + 4x + 1$$

has at exactly one real root.

Solution

Note that $f'(x) = 3x^2 + 4 > 0$.

Suppose $f(x)$ has at least two roots at $x = r_1$ and $x = r_2$.

Warm-up problem

Problem

Show that the polynomial

$$f(x) = x^3 + 4x + 1$$

has at exactly one real root.

Solution

Note that $f'(x) = 3x^2 + 4 > 0$.

Suppose $f(x)$ has at least two roots at $x = r_1$ and $x = r_2$.

If $r_1 = r_2$, then $f'(r_1) = 0$, which we know is impossible.

Warm-up problem

Problem

Show that the polynomial

$$f(x) = x^3 + 4x + 1$$

has at exactly one real root.

Solution

Note that $f'(x) = 3x^2 + 4 > 0$.

Suppose $f(x)$ has at least two roots at $x = r_1$ and $x = r_2$.

If $r_1 = r_2$, then $f'(r_1) = 0$, which we know is impossible.

If $r_1 \neq r_2$, then **Rolle's Theorem** or the **Mean Value Theorem** implies that there is a point x between r_1 and r_2 with $f'(x) = 0$, which is also impossible.

Warm-up problem

Problem

Show that the polynomial

$$f(x) = x^3 + 4x + 1$$

has at exactly one real root.

Solution

Note that $f'(x) = 3x^2 + 4 > 0$.

Suppose $f(x)$ has at least two roots at $x = r_1$ and $x = r_2$.

If $r_1 = r_2$, then $f'(r_1) = 0$, which we know is impossible.

If $r_1 \neq r_2$, then **Rolle's Theorem** or the **Mean Value Theorem** implies that there is a point x between r_1 and r_2 with $f'(x) = 0$, which is also impossible.

Therefore $f(x)$ has no more than one root.

Problem

Suppose that $f(x)$ has five distinct real roots. Show that

$$f(x) + 6f'(x) + 12f''(x) + 8f'''(x)$$

has at least two distinct real roots.

Problem

Suppose that $f(x)$ has five distinct real roots. Show that

$$f(x) + 6f'(x) + 12f''(x) + 8f'''(x)$$

has at least two distinct real roots.

Hint: consider $e^{x/2}f(x)$

Solution

Let $g(x) = e^{x/2}f(x)$.

Solution

Let $g(x) = e^{x/2}f(x)$.

Then $g(x)$ has at least 5 distinct zeros.

Solution

Let $g(x) = e^{x/2}f(x)$.

Then $g(x)$ has at least 5 distinct zeros.

By Rolle's Theorem, $g'(x)$ has at least four distinct zeros.

Solution

Let $g(x) = e^{x/2}f(x)$.

Then $g(x)$ has at least 5 distinct zeros.

By Rolle's Theorem, $g'(x)$ has at least four distinct zeros.

By Rolle's Theorem, $g''(x)$ has at least three distinct zeros.

Solution

Let $g(x) = e^{x/2}f(x)$.

Then $g(x)$ has at least 5 distinct zeros.

By Rolle's Theorem, $g'(x)$ has at least four distinct zeros.

By Rolle's Theorem, $g''(x)$ has at least three distinct zeros.

By Rolle's Theorem, $g'''(x)$ has at least two distinct zeros.

Solution

Let $g(x) = e^{x/2}f(x)$.

Then $g(x)$ has at least 5 distinct zeros.

By Rolle's Theorem, $g'(x)$ has at least four distinct zeros.

By Rolle's Theorem, $g''(x)$ has at least three distinct zeros.

By Rolle's Theorem, $g'''(x)$ has at least two distinct zeros.

Solution

Let $g(x) = e^{x/2}f(x)$.

Then $g(x)$ has at least 5 distinct zeros.

By Rolle's Theorem, $g'(x)$ has at least four distinct zeros.

By Rolle's Theorem, $g''(x)$ has at least three distinct zeros.

By Rolle's Theorem, $g'''(x)$ has at least two distinct zeros.

$$\begin{aligned} g'''(x) &= e^{x/2} \left(\frac{1}{8}f(x) + \frac{6}{8}f'(x) + \frac{6}{4}f''(x) + f'''(x) \right) \\ &= \frac{1}{8}e^{x/2} (f(x) + 6f'(x) + 12f''(x) + 8f'''(x)) \end{aligned}$$

Solution

Let $g(x) = e^{x/2}f(x)$.

Then $g(x)$ has at least 5 distinct zeros.

By Rolle's Theorem, $g'(x)$ has at least four distinct zeros.

By Rolle's Theorem, $g''(x)$ has at least three distinct zeros.

By Rolle's Theorem, $g'''(x)$ has at least two distinct zeros.

$$\begin{aligned}g'''(x) &= e^{x/2}\left(\frac{1}{8}f(x) + \frac{6}{8}f'(x) + \frac{6}{4}f''(x) + f'''(x)\right) \\ &= \frac{1}{8}e^{x/2}(f(x) + 6f'(x) + 12f''(x) + 8f'''(x))\end{aligned}$$

$e^{x/2}$ is never zero, so the term in parentheses has two zeros!

Problem

Suppose that

$$f(x) = (x - m_1)(x - m_2)(x - m_3)(x - m_4)(x - m_5)$$

for some distinct integers $m_1, \dots, m_5$. If we choose the values of these integers carefully, how many of the coefficients of $f(x)$ can we force to be zero?

Problem

Suppose that

$$f(x) = (x - m_1)(x - m_2)(x - m_3)(x - m_4)(x - m_5)$$

for some distinct integers $m_1, \dots, m_5$. If we choose the values of these integers carefully, how many of the coefficients of $f(x)$ can we force to be zero?

Solution

Ideas:

- *make 0 a root*
- *make other roots come in $\pm$ pairs:*

$$f(x) = x(x-a)(x+a)(x-b)(x+b) = x(x^2-a^2)(x^2-b^2) = x^5 - (a^2+b^2)x^3 + a^2b^2x$$

Solution

Take a, b to be integers.

$$m_3 = -m_2 = a, \quad a_1 = -a_4 = b$$

$$f(x) = x^5 - (a^2 + b^2)x^4 + a^2b^2x$$

Solution

Take a, b to be integers.

$$m_3 = -m_2 = a, \quad a_1 = -a_4 = b$$

$$f(x) = x^5 - (a^2 + b^2)x^4 + a^2b^2x$$

Suspicion: *the answer is we can make two coefficients zero.*

Solution

Take a, b to be integers.

$$m_3 = -m_2 = a, \quad a_1 = -a_4 = b$$

$$f(x) = x^5 - (a^2 + b^2)x^4 + a^2b^2x$$

Suspicion: *the answer is we can make two coefficients zero.*

Question: *can we have even more be zero???*

Solution

Solution

- *Just one nonzero coefficient?*

Solution

- *Just one nonzero coefficient?*

$f(x) = x^5$ has repeated roots...impossible!

Solution

- *Just one nonzero coefficient?*

$f(x) = x^5$ has repeated roots...impossible!

- *Just two nonzero coefficients?*

Solution

- *Just one nonzero coefficient?*

$f(x) = x^5$ has repeated roots...impossible!

- *Just two nonzero coefficients?*

$f(x) = x^5 + ax^n, n \geq 2$ has repeated roots...impossible!

Solution

- *Just one nonzero coefficient?*

$f(x) = x^5$ has repeated roots...impossible!

- *Just two nonzero coefficients?*

$f(x) = x^5 + ax^n$, $n \geq 2$ has repeated roots...impossible!

$f(x) = x^5 + ax$ has complex roots...impossible!

Solution

- *Just one nonzero coefficient?*

$f(x) = x^5$ has repeated roots...impossible!

- *Just two nonzero coefficients?*

$f(x) = x^5 + ax^n, n \geq 2$ has repeated roots...impossible!

$f(x) = x^5 + ax$ has complex roots...impossible!

$f(x) = x^5 + a$ has complex roots...impossible!

Problem

Suppose that $f(x) = x^3 + ax^2 + bx + c$ has three real roots r_1, r_2, r_3 with $r_1 \leq r_2 \leq r_3$. Show that

$$\sqrt{a^2 - 3b} \leq r_3 - r_1 \leq \frac{2}{\sqrt{3}} \sqrt{a^2 - 3b}$$

Solution

Solution

$$\begin{aligned}f(x) &= x^3 + ax^2 + bx + c \\&= (x - r_1)(x - r_2)(x - r_3) \\&= x^3 - (r_1 + r_2 + r_3)x^2 + (r_1r_2 + r_1r_3 + r_2r_3)x - r_1r_2r_3.\end{aligned}$$

Solution

$$\begin{aligned}f(x) &= x^3 + ax^2 + bx + c \\&= (x - r_1)(x - r_2)(x - r_3) \\&= x^3 - (r_1 + r_2 + r_3)x^2 + (r_1r_2 + r_1r_3 + r_2r_3)x - r_1r_2r_3.\end{aligned}$$

From this we see

$$\begin{aligned}a^2 - 3b &= r_1^2 + r_2^2 + r_3^2 - (r_1r_2 + r_1r_3 + r_2r_3) \\&= \frac{1}{2}(r_2 - r_1)^2 + \frac{1}{2}(r_3 - r_2)^2 + \frac{1}{2}(r_3 - r_1)^2 \\&= (r_3 - r_1)^2 - (r_3 - r_2)(r_2 - r_1)\end{aligned}$$

Solution

The **AM-GM inequality** says:

$$\sqrt{AB} \leq \frac{A+B}{2}.$$

Solution

The **AM-GM inequality** says:

$$\sqrt{AB} \leq \frac{A+B}{2}.$$

Therefore

$$0 \leq (r_3 - r_2)(r_2 - r_1) \leq \left(\frac{(r_3 - r_2) + (r_2 - r_1)}{2} \right)^2 = \left(\frac{r_3 - r_1}{2} \right)^2$$

Solution

The **AM-GM inequality** says:

$$\sqrt{AB} \leq \frac{A+B}{2}.$$

Therefore

$$\begin{aligned} 0 \leq (r_3 - r_2)(r_2 - r_1) &\leq \left(\frac{(r_3 - r_2) + (r_2 - r_1)}{2} \right)^2 = \left(\frac{r_3 - r_1}{2} \right)^2 \\ &\quad - \left(\frac{r_3 - r_1}{2} \right)^2 \leq -(r_3 - r_2)(r_2 - r_1) \leq 0 \end{aligned}$$

Solution

The **AM-GM inequality** says:

$$\sqrt{AB} \leq \frac{A+B}{2}.$$

Therefore

$$0 \leq (r_3 - r_2)(r_2 - r_1) \leq \left(\frac{(r_3 - r_2) + (r_2 - r_1)}{2} \right)^2 = \left(\frac{r_3 - r_1}{2} \right)^2$$

$$-\left(\frac{r_3 - r_1}{2} \right)^2 \leq -(r_3 - r_2)(r_2 - r_1) \leq 0$$

$$\frac{3}{4}(r_3 - r_1)^2 \leq (r_3 - r_1)^2 - (r_3 - r_2)(r_2 - r_1) \leq (r_3 - r_1)^2$$

Solution

Since

$$a^2 - 3b = (r_3 - r_1)^2 - (r_3 - r_2)(r_2 - r_1)$$

Solution

Since

$$a^2 - 3b = (r_3 - r_1)^2 - (r_3 - r_2)(r_2 - r_1)$$

This gives

$$\frac{3}{4}(r_3 - r_1)^2 \leq a^2 - 3b \leq (r_3 - r_1)^2$$

Solution

Since

$$a^2 - 3b = (r_3 - r_1)^2 - (r_3 - r_2)(r_2 - r_1)$$

This gives

$$\frac{3}{4}(r_3 - r_1)^2 \leq a^2 - 3b \leq (r_3 - r_1)^2$$

Therefore

$$\sqrt{a^2 - 3b} \leq r_3 - r_1,$$

Solution

Since

$$a^2 - 3b = (r_3 - r_1)^2 - (r_3 - r_2)(r_2 - r_1)$$

This gives

$$\frac{3}{4}(r_3 - r_1)^2 \leq a^2 - 3b \leq (r_3 - r_1)^2$$

Therefore

$$\sqrt{a^2 - 3b} \leq r_3 - r_1,$$

and also

$$r_3 - r_1 \leq \frac{2}{\sqrt{3}} \sqrt{a^2 - 3b}.$$